Marcos Gutierrez

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EDUCATION

University of California Santa Barbara, *Mechanical Engineering - GPA*: 3.47 Sep 2023 – Jun 2027 | Santa Barbara

Relevant Coursework:

Statics · Mechanics of Materials · Dynamics · Thermodynamics · Fluid Mechanics · Vibrations · Dynamical Systems · Heat Transfer · Finite Element Analysis (FEA) · Control System Design · Mechatronics

PROFESSIONAL EXPERIENCE

TA Aerospace (TransDigm Group), *Manufacturing Engineer Intern* Jun 2026 – Sep 2026 | Valencia, CA

- Diagnosed and repaired a large Vertical Lift Module (VLM) critical to production mold storage, reducing downtime from approximately 1 week to 2 days and avoiding up to $10,000 in service costs; developed standardized troubleshooting work instructions to improve future fault diagnosis and maintenance response
- Applied the Lean A3 problem-solving methodology to improve clamp assembly production output by analyzing current conditions, target output, root causes, countermeasures, implementation plans, and follow-up metrics.
- Developed a production management and accountability system for a high-value manufacturing area, establishing clear roles, responsibilities, escalation paths, and actions for assemblers, leads, supervisors, and manufacturing engineers.
- Designed visual production-management tools—including pitch charts, operator feedback systems, manual lead tracking boards, and quarterly production plotters—to track and communicate daily, weekly, and monthly manufacturing performance.
- Created cross-functional communication and escalation workflows involving manufacturing, quality, operations, and executive leadership to identify production constraints, assign corrective actions, and adjust production flow.
- Specified and supported the implementation of a Vorne XL Touch production monitoring system with industrial sensors to enable automated production counting, improve shop-floor visibility, and support data-driven manufacturing improvements.
- Designed modular fixtures and operator-support tooling in CATIA and produced engineering drawings for machining, improving assembly repeatability, operator efficiency, and ease of use.

CNSI - University of California, Santa Barbara, *Innovation Workshop Intern* Jan 2026 – Present | Santa Barbara, CA

- Support prototyping and fabrication workflows for researchers, students, and external partners across CNC machining, waterjet cutting, laser cutting, and additive manufacturing
- Operate manual mills, lathes, and CNC equipment to fabricate precision components within specified tolerances
- Produce rapid prototypes using FDM and resin-based 3D printing, optimizing print orientation, support structures, and post-processing for functional parts
- Operate laser cutting systems for precision 2D fabrication of structural and enclosure components
- Translate CAD models into manufacturing-ready formats, preparing toolpaths and machine setups to support iterative design cycles
- Promote safe equipment operation and assist with user training, machine setup, and troubleshooting in a shared advanced manufacturing environment

Gutierrez Family Landscaping, *Operations & Field Technician* Jan 2014 – present | Los Angeles, CA

- Supported operations of a family-owned landscaping business servicing 70+ residential properties
- Installed, repaired, and troubleshot irrigation and mechanical fluid systems, developing hands-on diagnostic skills
- Operated and maintained landscaping equipment and small engine systems
- Communicated directly with clients in English and Spanish, building long-term customer relationships
- Managed scheduling, logistics, and on-site problem solving in dynamic environments

PROJECTS

Gaucho Rocket Project, *Mechanical Engineer* Jun 2025 – Present

- Design and analyze structural components for the rocket airframe and landing leg deployment system
- Perform MIG welding and structural fabrication for landing leg assemblies, ensuring alignment, weld integrity, and load-bearing performance under impact conditions
- Conduct Finite Element Analysis (static and modal) on structural components including baseplates, brackets, and landing leg joints to evaluate stresses, deflection, and safety factors
- Designed and iterated the primary airframe structure in Fusion 360, incorporating landing leg assemblies, TVC mounting interfaces, and subsystem hardpoints to maintain structural continuity and alignment across integrated systems
- Refine designs through analysis-driven iteration and cross-subteam collaboration to improve stiffness, weight efficiency, and manufacturability
- Assist with technical documentation and design reviews, communicating analysis results and design intent

Fishing Line Kelp Desnagger, *Junior Capstone Design Project* Mar 2026 – Jun 2026

- Collaborated in a 4-person team to design and prototype a motor-driven kelp desnagger that travels along a fishing line to free snagged lures from kelp.
- Designed and integrated the device's circuit system, soldered a custom PCB, and supported motor control, sensor input, and power delivery for prototype operation.
- Performed torque analysis and prototype testing to evaluate motor capability, line-tension performance, and stall limits while working within a $200 project budget.
- Presented the final prototype at a student design fair, explaining the design problem, engineering analysis, testing process, and project results.

SKILLS

SolidWorks • Fusion 360 • ANSYS • CNC Machining • Soldering • GD&T Interpretation • MIG Welding • Mechanical Fabrication & Assembly • MATLAB • Microsoft Office • Mechanical Repair • Mechanical Troubleshooting • CATIA • Engineering Drawings • Fixture & Tooling Design • Lean A3 / Problem Solving • Production Data Tracking

LANGUAGES

Fluent in Spanish